

ZHAOYANG CHU

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RESEARCH INTERESTS

My research interests lie at the intersection of software engineering and artificial intelligence, with an emphasis on **coding agents**: how to build, evaluate, and make them reliable on real-world tasks.

EDUCATION

University College London

Ph.D. in Computer Science

Co-Supervised by Dr. He Ye and Prof. Federica Sarro

Sep 2025 – Sep 2028 (anticipated)

Huazhong University of Science and Technology

M.E. in Computer Science and Technology (Graduated with Honors)

Supervised by Prof. Yao Wan

Sep 2022 – Jun 2025

Huazhong Agricultural University

B.E. in Data Science and Big Data Technology (Graduated with Honors)

GPA: 3.93/4.0

Sep 2018 – Jun 2022

RESEARCH PAPERS

* indicates equal contribution. † indicates the corresponding author.

- [1] TerminalWorld: Benchmarking Agents on Real-World Terminal Tasks.

Zhaoyang Chu, Jiarui Hu*, Xingyu Jiang*, Pengyu Zou*, Han Li, Chao Peng, Peter O'Hearn, Earl T. Barr, Mark Harman, Federica Sarro, He Ye†.

NeurIPS 2026. [\[Paper\]](#) [\[Code\]](#) [\[Data\]](#) [\[Website\]](#)

26K+ Hugging Face Downloads 

- [2] OpenRUA: Robot-Use Agents Are Zero-Shot Visuomotor Policies.

Zhaoyang Chu, Earl T. Barr, Claire Le Goues, Peter O'Hearn, Mark Harman, Federica Sarro, He Ye†.

Under Review at ICLR 2027. [\[Paper\]](#) [\[Code\]](#)

- [3] Built for Humans, Not yet for Agents: Turning Apps into Agent-Native CLIs at Scale.

Ziyuan Zhang*, **Zhaoyang Chu***, Xing Hu†, Shanping Li, Xin Xia, Federica Sarro, He Ye.

Under Review at ICLR 2027.

- [4] ContextBench: A Benchmark for Context Retrieval in Coding Agents.

Han Li, Letian Zhu*, Bohan Zhang*, Rili Feng*, Jiaming Wang, Yue Pan, Earl T. Barr, Federica Sarro, **Zhaoyang Chu**†, He Ye†.

Under Review at ICLR 2027. [\[Paper\]](#) [\[Code\]](#) [\[Data\]](#) [\[Website\]](#)

- [5] Prometheus: Towards Long-Horizon Codebase Navigation for Repository-Level Problem Solving.

Yue Pan*, Zimin Chen*, Siyu Lu, **Zhaoyang Chu**, Xiang Li, Han Li, Yang Feng, Claire Le Goues, Federica Sarro, Martin Monperrus, He Ye†.

Under Review at ICSE 2027. [\[Paper\]](#) [\[Code\]](#)

1.1K+ GitHub Stars ★

- [6] An Iterative Test-and-Repair Framework for Competitive Code Generation.

Lingxiao Tang, Muyang Ye, **Zhaoyang Chu**, Xiaoxue Ren, Zhongxin Liu, Lingfeng Bao†, He Ye.

Under Review at ICSE 2027. [\[Paper\]](#)

- [7] HEAT: Automated Headless Testing for Desktop Applications.

Ziyuan Zhang, **Zhaoyang Chu**, Qi Zhan, Xing Hu†, Shanping Li, Xin Xia, He Ye.

Under Review at FSE 2027.

- [8] MT-GUI2CLI: Metamorphic-Test-Guided Semantic Migration from GUI Workflows to Reliable CLI Agents.

Shufan Wang, **Zhaoyang Chu**, Ziyuan Zhang, Xing Hu†, Xin Xia, Federica Sarro, He Ye.

Under Review at FSE 2027.

- [9] Scrub It Out! Erasing Sensitive Memorization in Code Language Models via Machine Unlearning.
Zhaoyang Chu, Yao Wan†, Zhikun Zhang, Di Wang, Zhou Yang, Hongyu Zhang, Pan Zhou, Xuanhua Shi, Hai Jin, David Lo.
ICSE 2026. [\[Paper\]](#) [\[Code\]](#)
- [10] Hallucinations in LLM-based Code Summarization: Unveiling, Detection, and Mitigation.
Guanghua Wan, Yuanning Feng, Yao Wan†, **Zhaoyang Chu**, Zhangqian Bi, Junxiao Han, Zhou Zhao, Hongyu Zhang, Pingpeng Yuan, Xuanhua Shi, Hai Jin.
FSE 2026. [\[Paper\]](#) [\[Code\]](#)
- [11] ExecVerify: White-Box RL with Verifiable Stepwise Rewards for Code Execution Reasoning.
Lingxiao Tang, He Ye, **Zhaoyang Chu**, Muyang Ye, Zhongxin Liu, Xiaoxue Ren, Lingfeng Bao†.
ACL 2026. [\[Paper\]](#) [\[Code\]](#)
- [12] CGBridge: Bridging Code Graphs and Large Language Models for Better Structure-Aware Code Understanding.
Zeqi Chen, **Zhaoyang Chu**, Yi Gui, Feng Guo, Yao Wan, Chuan Shi†.
ACL 2026 Findings. [\[Paper\]](#) [\[Code\]](#)
- [13] CODESYNC: Synchronizing Large Language Models with Dynamic Code Evolution at Scale.
Chenlong Wang*, **Zhaoyang Chu***, Zhengxiang Cheng*, Xuyi Yang, Kaiyue Qiu, Yao Wan†, Zhou Zhao, Xuanhua Shi, Dongping Chen.
ICML 2025. [\[Paper\]](#) [\[Code\]](#)
- [14] How to Select Pre-Trained Code Models for Reuse? A Learning Perspective.
Zhangqian Bi, Yao Wan†, **Zhaoyang Chu**, Yufei Hu, Junyi Zhang, Hongyu Zhang, Guandong Xu, Hai Jin.
SANER 2025. [\[Paper\]](#) [\[Code\]](#)
IEEE TCSE Distinguished Paper Award 🏆
- [15] Can Large Language Models Serve as Evaluators for Code Summarization?
Yang Wu, Yao Wan†, **Zhaoyang Chu**, Wenting Zhao, Ye Liu, Hongyu Zhang, Xuanhua Shi, Philip S. Yu.
TSE 2025. [\[Paper\]](#) [\[Code\]](#)
- [16] Wait, We Don't Need to "Wait"! Removing Thinking Tokens Improves Reasoning Efficiency.
Chenlong Wang, Yuanning Feng, Dongping Chen, **Zhaoyang Chu**, Ranjay Krishna†, Tianyi Zhou†.
EMNLP 2025 Findings. [\[Paper\]](#)
- [17] TESTEVAL: Benchmarking Large Language Models for Test Case Generation.
Wenhan Wang*, Chenyuan Yang*, Zhijie Wang*, Yuheng Huang, **Zhaoyang Chu**, Da Song, Lingming Zhang, An Ran Chen, Lei Ma.
NAACL 2025 Findings. [\[Paper\]](#) [\[Code\]](#) [\[Website\]](#)
- [18] Graph Neural Networks for Vulnerability Detection: A Counterfactual Explanation.
Zhaoyang Chu, Yao Wan†, Qian Li, Yang Wu, Hongyu Zhang, Yulei Sui, Guandong Xu, Hai Jin.
ISSTA 2024. [\[Paper\]](#) [\[Code\]](#)

INTERNSHIPS & VISITS

University of Illinois Urbana-Champaign
Research Intern

Supervised by Prof. Lingming Zhang
May 2024 – Oct 2024

HONORS & AWARDS

IEEE TCSE Distinguished Paper Award

2025